1. $y = \frac{(x^2+1)^{\sin x}}{e^{x \cos x}}$ find y'

$y' = (x^2+1)^{\sin x} e^{-x \cos x}$

$f := (x^2+1)^{\sin x} \quad g := e^{-x \cos x}$

$\ln f = \sin x \ln(x^2+1) \quad g' = (-\cos x + x \sin x) e^{-x \cos x}$

$\frac{f'}{f} = \cos x \ln(x^2+1) + \sin x \frac{2x}{x^2+1}$

$f' = \left(\cos x \ln(x^2+1) + \frac{2x \sin x}{x^2+1}\right) f$

$y' = f'g + fg'$

$y' = \left(\cos x \ln(x^2+1) + \frac{2x \sin x}{x^2+1}\right) f e^{-x \cos x} + (-\cos x + x \sin x) e^{-x \cos x} f$

$y' = e^{-x \cos x} (x^2+1)^{\sin x} \left(\left(\cos x \ln(x^2+1) + \frac{2x \sin x}{x^2+1}\right) + (-\cos x + x \sin x)\right)$

2. $y = \ln \frac{\sqrt{1+x^2}}{1+\sin x}$ find y'

$f := \frac{\sqrt{1+x^2}}{1+\sin x}$

$y = \ln f$

$y' = \frac{f'}{f}$

$f' = \frac{x \frac{1+\sin x}{\sqrt{1+x^2}} - \sqrt{1+x^2} \cos x}{(1+\sin x)^2}$

$y' = \frac{\frac{x \frac{1+\sin x}{\sqrt{1+x^2}} - \sqrt{1+x^2} \cos x}{(1+\sin x)^2}}{\frac{\sqrt{1+x^2}}{1+\sin x}}$

$y' = \frac{x \frac{1+\sin x}{\sqrt{1+x^2}} - \sqrt{1+x^2} \cos x}{(1+\sin x)\sqrt{1+x^2}}$

$y' = \frac{x}{1+x^2} - \frac{\cos x}{1+\sin x}$

3. $y = \arctan\left(\frac{2x}{1-x^2}\right)$ find y'

$f := \frac{2x}{1-x^2}$

$y = \arctan f$

$y' = \frac{f'}{1+f^2}$

$f' = \frac{2(1-x^2)+4x^2}{(1-x^2)^2}$

$f' = \frac{2-2x^2+4x^2}{(1-x^2)^2}$

$f' = 2\frac{x^2+1}{(1-x^2)^2}$

$y' = 2\frac{\frac{x^2+1}{(1-x^2)^2}}{1+\left(\frac{2x}{1-x^2}\right)^2}$

$y' = 2\frac{x^2+1}{(1-x^2)^2+4x^2}$

$y' = 2\frac{x^2+1}{1-2x^2+x^4+4x^2}$

$y' = 2\frac{x^2+1}{x^4+2x^2+1}$

$y' = 2\frac{x^2+1}{(x^2+1)^2}$

$y' = \frac{2}{x^2+1}$